Fixed-target conservation and its contact residuals

Research audit note

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1 Fixed-target contact and conservation with explicit residuals

The purpose of this note is to close the contact-to-conservation implication, including its endpoint conventions. It also supplies an identity away from contact, which can be checked on arbitrary finite laws. It establishes no numerical acquisition schedule or final exclusion. Throughout, sums have independent identically distributed summands with mean zero, variance one and finite third absolute moment ρ. Fix $n \geq 2$, $c > 0$, and a real raw sum threshold H. Set

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \leq H), \quad a = \sqrt{n}\{F - \Phi(z)\} - c\rho, \quad \varphi = \phi(z), \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3.$$

In particular a is not divided by ρ.

1.1 Attainment, contamination and the entire closed support

Proposition 1.1. *For each fixed n and $c > 0$, the supremum of a over these laws and thresholds is attained, and its attaining discrepancy is positive. At every maximizing pair,*

$$\begin{aligned} I(y) = n^{3/2}\{h(y) - F\} + ny\varphi + \tfrac{1}{2}H\varphi(y^2 - 1) \\ + \tfrac{3}{2}c\rho(y^2 - 1) - c\{|y|^3 - \rho - 3y\beta\} \leq 0, \qquad h(y) = \mathbb{P}(S_{n-1} \leq H - y), \end{aligned} \tag{1}$$

with equality on the whole closed support. That support is compact. Neither statement assumes $a \geq 0$ or $H \geq 0$.

Proof. Write A for the supremum. Since $\rho \geq 1$ and the discrepancy is at most one, $A \leq \sqrt{n} - c$. Symmetric Bernoulli evaluated at $H = n$ gives $A \geq \sqrt{n}(1 - \Phi(\sqrt{n})) - c > -c$. For a maximizing sequence, $a_j \to A$ and $c\rho_j \leq \sqrt{n} - a_j$ give a common third-moment ceiling. The discrepancies satisfy $\sqrt{n}(F_j - \Phi(H_j/\sqrt{n})) = a_j + c\rho_j \geq a_j + c > \eta$ eventually for some $\eta > 0$. Chebyshev bounds the negative threshold tail by n/H_j^2; the positive discrepancy is at most the normal tail, also at most n/H_j^2. Thus H_j is bounded.

Tightness gives a subsequence $P_j \Rightarrow P$, $H_j \to H$ and $\rho_j \to b$. The common third-moment ceiling M gives tail bounds $\int_{|x|>K} x^2 dP_j \leq M/K$ and $\int_{|x|>K} |x| dP_j \leq M/K^2$. Bounded continuous truncations therefore preserve mean and variance. Lower semicontinuity gives $1 \leq \rho(P) \leq b$. For fixed n, the sum laws converge weakly. Applying the closed-set inequality first at $H + t$ and then letting $t \downarrow 0$ gives $\limsup F_j(H_j) \leq F_P(H)$, including an atom at H. Consequently

$$a(P, H) \geq A + c\{b - \rho(P)\} \geq A.$$

Maximality forces equality and $\rho(P) = b$. Convergence of third moments is a conclusion here; it was not inferred from bounded third moments. The discrepancy remains positive since $A + c\rho \geq A + c > 0$.

For contact, contaminate the raw marginal by $(1-\epsilon)P+\epsilon\delta_y$. Its mean is ϵy, variance $v_\epsilon = 1+\epsilon(y^2-1)-\epsilon^2y^2$, and centered third absolute moment has derivative $|y|^3-\rho-3y\beta$ at zero. The last assertion follows by dominated differentiation of $|x-\epsilon y|^3$, with bound $3|y|(|x|+|y|)^2$. Evaluate the standardized law at the standardized image of the fixed raw event $S_n \le H$. The raw event probability is a polynomial in ϵ, with derivative $n(h(y)-F)$ even at an atom collision. The normal argument is $(H-n\epsilon y)/\sqrt{nv_\epsilon}$, whose derivative is $-\sqrt{n}y-H(y^2-1)/(2\sqrt{n})$. The one-sided derivative of the excess is exactly (1); maximality makes it nonpositive.

Its integral against P is zero. All terms are integrable. The closed-event map h is upper semicontinuous; at an atom it takes the larger one-sided value. Thus I is upper semicontinuous. Strict negativity at a closed-support point would hold with a uniform negative margin on a neighborhood of positive P-mass, contradicting its zero integral. Finally $I(y) \le -c|y|^3+C_2y^2+C_1|y|+C_0$ for constants depending on this pair. It is strictly negative on both sufficiently distant rays, proving compact support. No higher moments were used before this step. □

The sign of the attained maximum detects the corresponding row failure, including equality. If C_n denotes the supremum of the normalized absolute discrepancy, reflection gives, including an atom at H,

$$\mathbb{P}(-S_n \le -H)-\Phi(-z) = \Phi(z)-\mathbb{P}(S_n < H) \ge \Phi(z)-F.$$

Thus each negative closed-CDF error is bounded by a positive closed-CDF error of the reflected law; equality at an atom is not assumed. Consequently C_n is also the supremum of positive closed scores r. Since $a=\rho(r-c)$, $C_n<c$ gives $A \le C_n-c<0$, and $C_n>c$ gives $A>0$. If $C_n=c$, choose $r_j \uparrow c$ with $r_j \ge c/2$. Then $\rho_j \le 2\sqrt{n}/c$, so $\rho_j(r_j-c)\to 0$ and $A=0$. Hence a weak failure $C_n \ge c$ supplies an attained pair with $a \ge 0$. This does not supply any finite base or exclude that pair.

Equivalently, $\Delta = Q-h \ge 0$ everywhere and $\Delta=0$ on the closed support, where

$$Q(y) = F+\frac{H\varphi+c\rho}{2n^{3/2}}-\frac{n\varphi+3c\beta}{n^{3/2}}y-\frac{H\varphi+3c\rho}{2n^{3/2}}y^2+\frac{c}{n^{3/2}}|y|^3. \tag{2}$$

1.2 An identity that exposes both contact uses

We now allow an arbitrary compactly supported standardized law, retaining the definitions of a, Q, h, Δ above without assuming contact. Put $\nu = \mathcal{L}(H-S_{n-1})$ and $\tau(t)=\mathbb{E}[X\mathbf{1}_{X\ge t}]$. Then $\tau \ge 0$, $\int \tau = 1$. Over the bounded complementary intervals (u,v) of the closed support K, define

$$J=\sum \tau_{u,v}(v-u)^2, \qquad J_1=\sum \tau_{u,v}\int_u^v (t-u)t\,dt,$$
$$J_2=\sum \tau_{u,v}\int_u^v (t-u)t|t|\,dt, \qquad L=\sum \tau_{u,v}\int_{[u,v)}(t-u)\,d\nu(t).$$

Here $\tau_{u,v}$ is the constant interior value. Introduce two contact residuals:

$$\xi=\mathbb{E}[X\Delta(X)], \qquad B=\int_K \tau(t)\Delta(t)\,dt+\sum_{(u,v)}\tau_{u,v}(v-u)\Delta(v). \tag{3}$$

The integral over K is a Lebesgue integral, not an expectation under P. It must be retained when the closed support has positive Lebesgue measure. If $K \subset [-R,R]$, then $J \le 2R$, $|J_1| \le RJ/2$, $|J_2| \le R^2J/2$, and $0 \le L \le 2R$. Also Δ is bounded on the convex hull, so the sum in B is absolutely convergent because $\sum \tau_{u,v}(v-u) \le 1$.

Let $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$, and put

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$

$$U = J_1 + (M_4 - 3)/6, \quad \varepsilon = \mathbb{E}q_z(S_n/\sqrt{n}) - \mathbb{E}q_z(Z_0) - 2\gamma\varphi/(3\sqrt{n}), \quad q_z(x) = (x - z)|x - z|,$$

where Z_0 is standard normal. Define $E_n = -n^{3/2}\varepsilon/2$ and

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \varphi U) - H^2(c\rho + \gamma\varphi/2) + E_n. \tag{4}$$

Theorem 1.2 (Conservation with contact residuals)**.** *For every such law, every $H \in \mathbb{R}$ and every $c > 0$,*

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c + n^{3/2}(B - H\xi). \tag{5}$$

At a fixed-target maximizing pair, $B = \xi = 0$. Hence the exact fixed-target master identity follows, with coefficient $n + H^2$.

Proof. There are two direct evaluations of $\mathcal{A} = \int \tau\Delta$. Since $h(t) = \nu([t, \infty))$, one has $h(t) = h(v) + \nu([t, v))$ inside a gap. Thus bounded Fubini gives

$$\int_u^v \Delta(t)dt = (v - u)\Delta(v) + \int_u^v (t - u)(-Q'(t))dt - \int_{[u,v)} (t - u)d\nu(t).$$

An atom at v belongs to $h(v)$ and is not in the last integral; an atom at u is included but multiplied by zero. Summing, including the support integral in (3), gives

$$\mathcal{A} = \frac{\varphi J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\varphi + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L + B. \tag{6}$$

For the second evaluation, signwise Fubini proves $\int \tau g' = \mathbb{E}[Xg(X)]$ for absolutely continuous g with bounded derivative on the relevant compact range. In particular

$$\int t\tau = \gamma/2, \quad \int t^2\tau = M_4/3, \quad \int |t|^3\tau = M_5/4.$$

Apply that same identity conditionally to $g_s(x) = \min(s + x, H)$ and then use exchangeability. The choice of derivative at its single corner is immaterial under the density τ. It follows directly that

$$\int \tau(t)h(t)dt = \mathbb{E}[X_n \min(S_n, H)] = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T,$$

where $T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{Z \le z}]$, $Z = S_n/\sqrt{n}$. The last expression vanishes at an endpoint atom. Integrating (2) against τ therefore yields

$$\mathcal{A} = F - \frac{\gamma\varphi}{2\sqrt{n}} + \frac{H\varphi(3 - M_4)/6 + cK_q}{n^{3/2}} - T. \tag{7}$$

Set $M = \mathbb{E}[Z\mathbf{1}_{Z \le z}]$. Since $\mathbb{E}[Xh(X)] = M/\sqrt{n}$, integration of $XQ(X)$ against P gives

$$M + \varphi = -\frac{\gamma H\varphi}{2n} + \frac{cK_1}{n} - \sqrt{n}\xi. \tag{8}$$

The pointwise identity $q_z(x) = (x - z)^2 - 2(x - z)^2\mathbf{1}_{x \le z}$, with the same identity for a normal variable, gives

$$T - \Phi(z) = -\varepsilon/2 - \gamma\varphi/(3\sqrt{n}) + z(M + \varphi) - z^2(F - \Phi(z)).$$

Inserting (8) and $\sqrt{n}(F-\Phi(z)) = c\rho + a$ into (7) gives

$$\sqrt{n}\mathcal{A} = c\rho + a - \gamma\varphi/6 + P_0/n + \sqrt{n}\varepsilon/2 + H\sqrt{n}\xi,$$
$$P_0 = H^2(c\rho + a + \gamma\varphi/2) - H\{cK_1 + \varphi(M_4 - 3)/6\} + cK_q.$$

Equate this with (6) multiplied by $\sqrt{n}$, then multiply by n and collect terms. This is (5). Under closed-support contact, every term defining B and ξ vanishes. □

The compensation/Fourier theorem already proved in the v7 note applies to this same E_n; its comparator has transform $e^{-nt^2/2}(1 - in\gamma t^3/6)$. It is not a new contact premise. The residual identity above also explains why testing the residual-free formula on arbitrary nonmaximizing laws would be an invalid test. For example, let X be uniform on $[-\sqrt{3}, \sqrt{3}]$ and take $H = 0$. There are no support gaps, $F = T = 1/2$, $\beta = \gamma = 0$, and $K_q = 3\sqrt{3}/40$. Therefore $B = \int_K \tau\Delta = cK_q/n^{3/2} > 0$. Omitting the support integral would give a false identity, even though the gap sums are all empty. This is an off-contact negative control, not a counterexample to the contact theorem.

1.3 Defect positivity and the first resource license

Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $\sigma = \sqrt{10}$. Signwise Cauchy gives $a_\pm^2 \le mb_\pm$ and $m^2 \le a_+a_-$, where $a_\pm = \mathbb{E}X_\pm^2$ and $b_\pm = \mathbb{E}X_\pm^3$. Therefore $b_+b_- \ge m^2$, or $\gamma^2 + 4m^2 \le \rho^2$. A gap on one side of zero has $(v-u)^2 \le 2\int_u^v |t|dt$. If a central gap $(-A, B)$ exists, its extra contribution is $2mAB$, and $0 \le \mathbb{E}[(X+A)(X-B)] = 1 - AB$. Signwise Tonelli then gives $J \le \rho + 2m$. It follows that

$$D = (3+\sigma)\rho - \gamma - 3J = 3(\rho + 2m - J) + (\sigma\rho - \gamma - 6m) \ge 0. \tag{9}$$

The last term is nonnegative by Cauchy in two dimensions. These statements hold with or without an atom at zero.

Specialize to $c = C_{\mathrm{E}} = (3+\sigma)\phi(0)/6$ and contact. With $w = H^2/(2n)$, $e = e^{-w}$, the identity becomes

$$\mathcal{R}_c = (n + H^2)a + nc\rho(1-e) + \frac{n\phi(0)e}{6}D + n^{3/2}L. \tag{10}$$

If $a \ge 0$, every term is nonnegative. This specialization to C_{E} is essential for this definition of D.

Corollary 1.3. *At such a contact pair with $a \ge 0$, assume $n \ge N \ge 2$, $\rho \ge l > 0$, $0 < c_- \le c$, and a proved raw bound $\mathcal{R}_c \le T < Nc_-l$. A negative T excludes the pair. Otherwise*

$$\gamma + 3J > 0, \qquad e \ge 1 - T/(Nc_-l) > 0,$$
$$\mathcal{U} := (n + H^2)a + n\phi(0)D/6 + n^{3/2}L \le \mathcal{R}_c,$$
$$\mathcal{U} \le \mathcal{R}_c - \tfrac{1}{2}(c_-l - T/N)H^2.$$

No condition $D < 2$ or own-score upper bound is required.

Proof. Set $M_c = c\rho - \phi(0)D/6 = \phi(0)(\gamma + 3J)/6$. The general master identity gives

$$eM_c = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \ge c\rho - T/n > 0.$$

Thus $M_c > 0$ and, since $D \ge 0$, $M_c \le c\rho$. The density bound follows. Regrouping (10) gives $\mathcal{R}_c = \mathcal{U} + n(1-e)M_c$, proving the unweighted bound. Finally

$$n(1-e)M_c = n(e^w - 1)eM_c \ge \frac{H^2}{2}(c\rho - T/n) \ge \frac{H^2}{2}(c_-l - T/N).$$

This prices one term once. It is a bound on $\mathcal{U}$, not an improved bound on the raw remainder. □

All conclusions are at each actual index and raw threshold. A common moment cell and common raw upper bound make them uniform even for varying maximizing laws. There is no exchange of a limit in n with contact, gaps, moments or an integral. The remaining application inputs are the quantitative support/moment bounds, complete Fourier envelope, ordered numerical acquisition and final exclusion, each at its own scope.